Exercise 8.5

1. $6x-7y-7$ 2. $-(x+20y+16)$ 3. $(2x-y)dx+\left(-x+\frac{1}{2}y\right)dy$

4. $(y+z)dx+(x+z)dy+(y+x)dz$

Exercise 8.6

1. $e^t\left(2e^t\sin t+3\sin^4 t+e^t\cos t+12\sin^3 t\cos t\right)$, 1

2. $\left(1+e^{2t}\right)^2\left[\cos^3 t\left(1+e^{2t}\right)-\sin t\sin 2t\left(1+e^{2t}\right)+6e^{2t}\sin t\cos^2 t\right]$

3. $4e^{2t}$ 4. $-e^{-2t}\left[\sin 2t-\cos 2t\right]$ 5. $18e^{3s}-3e^s\cos s+3e^s\sin s-4\sin s\cos s$, 15

6. $\frac{3e}{1+e^2}+2\tan^{-1}e$, $\frac{e}{1+e^2}$

7. $te^{st^2}\left[t\sin\left(s^2t\right)+2s\cos\left(s^2t\right)\right]$, $\frac{du}{dt}=se^{st^2}\left[2t\sin\left(s^2t\right)+s\cos\left(s^2t\right)\right]$, $e\left[\sin(1)+2\cos(1)\right]$, $e\left[2\sin(1)+\cos(1)\right]$

8. $3s^3(e^{3t}+s^2e^{-t})$, $3s^2e^t(e^{2t}-5e^{-2t}s^2)$ 9. $2u(1+2v)$, $2\left(u^2-v\right)$, 3, $\frac{-3}{2}$

Exercise 8.7

1. (i) not homogeneous (ii) Homogeneous, deg.3
   (iii) homogeneous, deg.0 (iv) not homogeneous 6. 5

Exercise 8.8

1	2	3	4	5	6	7	8
(2)	(2)	(2)	(4)	(3)	(2)	(4)	(2)
9	10	11	12	13	14	15	
(4)	(1)	(2)	(3)	(2)	(4)	(1)	

Chapter 9

Exercise 9.1

1. 0.6 2. 0.855 3. 0.375

Exercise 9.2

1. (i) $\frac{13}{2}$ (ii) $\frac{25}{3}$

Exercise 9.3

1. (i) $\frac{1}{4}\log\frac{5}{3}$ (ii) $\frac{\pi}{8}$ (iii) $\frac{\pi}{2}-1$ (iv) $e^{\frac{\pi}{2}}$ (v) $\frac{8}{21}$ (vi) $\frac{1}{2}$

2. (i) 0 (ii) π (iii) $\frac{\pi-2}{4}$ (iv) 0 (v) 0 (vi) $\frac{13}{10}$ (vii) $\frac{\pi}{4}$
   (viii) $\frac{\pi}{8}\log 2$ (ix) $\frac{\pi}{2}(\pi-2)$ (x) $\frac{\pi}{8}$ (xi) $\frac{\pi^2}{2}$

Exercise 9.4

1. $\frac{3}{8}-\frac{19}{8}e^{-2}$ 2. $\frac{1}{\sqrt{2}}\left(\frac{\pi}{12}+\frac{1}{9}\right)$ 3. $1+e^{\frac{\pi}{4}}\left[\frac{\pi}{4}-1\right]$ 4. $-\frac{\pi}{4}$

Exercise 9.5

1. (i) $\frac{\pi}{2\sqrt{6}}$ (ii). $\frac{\pi}{6\sqrt{5}}$

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